Question No 18

```
import pandas as pd
import numpy as np

data = {
    "Student_ID": [101, 102, 103, 104, 105, 105],
    "Name": ["Amit", "Riya", "Rahul", "Sneha", "Priya", "Priya"],
    "Marks": [85, np.nan, 78, 90, 88, 88],
    "Attendance": [90, 85, np.nan, 95, 88, 88],
    "Grade": ["A", "B", None, "A", "B", "B"]
}

df = pd.DataFrame(data)
df
```

	Student_ID	Name	Marks	Attendance	Grade
0	101	Amit	85.0	90.0	A
1	102	Riya	NaN	85.0	B
2	103	Rahul	78.0	NaN	None
3	104	Sneha	90.0	95.0	A
4	105	Priya	88.0	88.0	B
5	105	Priya	88.0	88.0	B

Next steps: Generate code with df New interactive sheet

Display Dataset Summary Including Missing Value Counts

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Student_ID  6 non-null      int64
 1   Name        6 non-null      object
 2   Marks       5 non-null      float64
 3   Attendance  5 non-null      float64
 4   Grade       5 non-null      object
dtypes: float64(2), int64(1), object(2)
memory usage: 372.0+ bytes
```

```
df.isnull().sum()
```

	0
Student_ID	0
Name	0
Marks	1
Attendance	1
Grade	1

dtype: int64

Replace Missing Attendance with Median of Attendance

```
median_attendance = df['Attendance'].median()
median_attendance
```

```
88.0
```

Remove Duplicate Student Records

```
df = df.drop_duplicates()
df
```

	Student_ID	Name	Marks	Attendance	Grade
0	101	Amit	85.0	90.0	A
1	102	Riya	NaN	85.0	B
2	103	Rahul	78.0	88.0	None
3	104	Sneha	90.0	95.0	A
4	105	Priya	88.0	88.0	B

Next steps: Generate code with df New interactive sheet

Display Records Where Attendance Is Greater Than 88

```
df[df['Attendance'] > 88]
```

	Student_ID	Name	Marks	Attendance	Grade
0	101	Amit	85.0	90.0	A
3	104	Sneha	90.0	95.0	A

Replace Grade "B" with "C"

```
df['Grade'] = df['Grade'].replace("B", "C")
df
```

```
/tmp/ipython-input-1559163977.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-ver
  df['Grade'] = df['Grade'].replace("B", "C")
```

	Student_ID	Name	Marks	Attendance	Grade
0	101	Amit	85.0	90.0	A
1	102	Riya	NaN	85.0	C
2	103	Rahul	78.0	88.0	None
3	104	Sneha	90.0	95.0	A
4	105	Priya	88.0	88.0	C

Next steps: Generate code with df New interactive sheet